

beaches around the world, together they contain around 7.5×10^{18} grains of sand. Thus, the addressing space of IPv6 is sufficient to give each grain of sand its own unique IP address—and to do this for approximately 5×10^{19} Earth-like worlds—so I don't think we're going to run out of IPv6 addresses in the foreseeable future."

Apart from this, there is also a class of low-power wireless personal area network (LoWPAN) protocols. In such networks, the nodes are very small and may be powered by an 8-bit, 16-bit or 32-bit microcontroller with 64kB or more of Flash memory. The most commonly used underlying transport protocol is defined by the IEEE 802.15.4 standard, and the packets of data are only 127 bytes in size.

Above this there is wireless stack, and then the set of applications to be run on the node. This is the kind of network that is normally used in home or industrial automation for the nodes to communicate with each other. Some of the technologies in this category include ZigBee, Z-Wave, EnOcean and SNAP. SNAP is the technology that has been chosen by Google for its Android@ Home project—a smart home-like effort.

So, we have standard Internet-level networks like TCP/IP on one side, and LoWPANs on the other. However, in order to move closer to the Internet of Things, we need all the nodes in the LoWPAN to also be Internet-capable and have their own Internet address—so that we can access and control these nodes from anywhere in the world using any device connected to the Internet (like a computer or a mobile phone).

This problem is not as easy as it sounds, because the size of the packets in the standard network can range from 1500 to 9000+ bytes, while the packets in the LoWPANs are only 127 bytes in size. LoWPANs also have lower data bandwidths and low power. As a result of some industry efforts, we now have an open standard called 6LoWPAN (where the '6' stands for IPv6), which describes how to take a large packet from a standard Internet-



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level network and strip it down by removing a lot of high-level complexity and redundancy, to make it small enough for handling by the LoWPAN.

The 6LoWPAN specification also describes how to take small LoWPAN packets and reformat them for passing on to a higher-level network. So, it is expected that we will soon see 6LoWPAN wireless protocol stacks. However, some challenges related to creating and deploying applications based on this standard are to be ironed out before this happens.

Keeping away the snoopers

When we speak of embedded Internet systems—be it in a mobile phone or in a television, security is a key issue. Especially, when you think about the context of an Internet of Things, where all ambient objects have computers embedded in them, and will be connected to the Internet and to other objects around them including your own computer, the points of 'break-in' also increase. In the worst case, your microwave or washing machine could serve as the unguarded back entrance for somebody to break into your computer and steal all your confidential data. Hence protection from malicious access is an important aspect of designing and developing

embedded Internet systems.

Sable shares an example, "With the power of embedded devices in the car infotainment space, hackers can take control over the system and remotely take control of your car. In some cases, the compromise of personal information can cause unwanted distress as well. The current-day embedded systems are a mixed bag when it comes to security; some handle it well, others don't."

There are such many situations where the security challenge deepens in the case of embedded systems. For example, where an Internet-enabled device needs to connect to an untrusted foreign network, say a wireless access point at an airport. The cryptographic algorithms should be powerful enough to allow sufficient security, while still having low power consumption and capable of running on systems with resource restrictions. "There are other scenarios where the security weakness could stem from the basic Internet protocol itself (say like HyperText Transfer Protocol (HTTP) streaming from a Web service in which a firewall cannot filter sensitive transmission) and needs additional layers of security," adds Mani.

While these sound similar to the security issues faced by personal computers and other larger systems, the chal-